Pb 1 Fie un sistem dat prin matricele:

$$A = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad C = [1 \ 0] \quad D = [0]$$

Se cere:

a) Structura sistemului

b) Verificati: b1) controlabilitate
b2) observabilitate
b3) stabilitate internă
b4) —"— —— externă

a) Ordin sistem = $\dim A = 2 \Rightarrow$ grad polinomul de la numitor
Fct de transfer = 2 $\Rightarrow$ există 2 poli $\Rightarrow$ 2 variabile de stare.

Nr intrări = 1 = nr. coloane B

Nr. ieșiri = 1 = nr. linii C

b1) $R_c = [B \ AB \ A^2B \ \dots \ A^{n-1}B]$

Dar $n = 2 \Rightarrow R_c = [B \ AB] = \begin{bmatrix} -2 & 2 \\ 1 & -1 \end{bmatrix}$ (B, AB)

$\begin{vmatrix} -2 & 2 \\ 1 & -1 \end{vmatrix} = 2 - 2 = 0 \Rightarrow \operatorname{rang} R_c \neq 2 \Rightarrow$ sistem NU este controlabil

b2) $R_o = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$ Dar $n = 2$

$\Rightarrow R_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \begin{matrix} \} C \\ \} CA \end{matrix}$

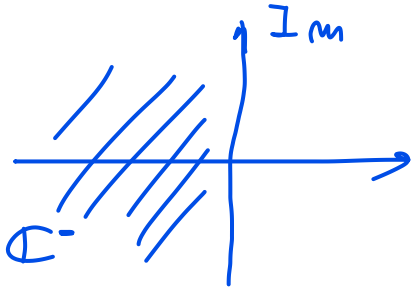
$\begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} = 0 + 0 = 0 \Rightarrow \operatorname{rang} R_o \neq 2 \Rightarrow$ sistem NU este observabil

b3) $|sI-A|=0 \Rightarrow s\cdot\begin{bmatrix}1&0\\0&1\end{bmatrix}-\begin{bmatrix}-1&0\\1&1\end{bmatrix}=\begin{bmatrix}s+1&0\\-1&s-1\end{bmatrix}$

$\Rightarrow |sI-A| = (s+1)(s-1)=0 \Rightarrow \Gamma(A)\in\{-1;1\}$. $\notin \mathbb{C}^-$

$\Rightarrow \Gamma(A)\notin \mathbb{C}^- \Rightarrow$ sistem

Nu este stabil intern



b4) $H(s)= C\cdot(sI-A)^{-1}\cdot B.$

$[sI-A]=\begin{bmatrix}s+1&0\\-1&s-1\end{bmatrix} \Rightarrow (sI-A)^T=\begin{bmatrix}s+1&-1\\0&s-1\end{bmatrix}$

$\det(sI-A)=(s+1)(s-1)$

$(sI-A)^{-1}=\frac{1}{\det}\begin{bmatrix}s-1&0\\+1&s+1\end{bmatrix}$

$H(s)=\frac{1}{(s+1)(s-1)}\begin{bmatrix}1&0\end{bmatrix}\cdot\begin{bmatrix}s-1&0\\1&s+1\end{bmatrix}\cdot\begin{bmatrix}-2\\1\end{bmatrix}=\frac{1}{(s+1)(s-1)}\begin{bmatrix}s-1&0\end{bmatrix}\begin{bmatrix}-2\\1\end{bmatrix}$

$=\frac{-2(s-1)}{(s-1)(s+1)} \rightarrow$ poli $\in\{-1;1\}\notin \mathbb{C}^- \Rightarrow$ sistem

Nu este stabil extern

Pr 2 Se dă sistemul cu funcția de transfer:

$$H(s) = \frac{s+2}{(s+1)(s+3)}$$

Se cere:

a) structura sistemului

b) o reprezentare în ss care să pună în evidență coeficienții polinomului caracteristic (numitor fct. transfer)

c) o repr. în ss care să pună în evidență polii sistemului

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a) Ordin Sistem = 2 = grad numitor fct transfer. 3

$\Rightarrow \exists$ 2 poli $\begin{matrix} -1 \\ -3 \end{matrix}$ $\Rightarrow$ matricea A are dim 2×2 $\Rightarrow$ 2 vare stare

Fct de transfer este raport de polinoame $\Rightarrow$ 1 intrare

$\Rightarrow$ B are 1 coloană

1 ieșire $\Rightarrow$

$\Rightarrow$ C are 1 linie

b) $H(s) = \frac{s+2}{(s+1)(s+3)} = \frac{s+2}{s^2+4s+3} = \frac{Y(s)}{U(s)}$

$\Rightarrow Y(s) = \frac{(s+2)U(s)}{s^2+4s+3}$ $\xrightarrow{\mathcal{L}^{-1}}$

$$\begin{cases} \dot{x} = A \cdot x + B \cdot u & (1) \\ y = C \cdot x + \Delta \cdot u & (2) \end{cases}$$

Notăm $Z(s) = \frac{U(s)}{s^2+4s+3} \Rightarrow Z(s) \cdot [s^2+4s+3] = U(s)$

$\mathcal{L}^{-1} \Rightarrow \boxed{\ddot{z} + 4\dot{z} + 3z = u}$

$\mathcal{L}[\ddot{x}] = s^2 X(s)$

$\mathcal{L}[\dot{x}] = s^1 X(s)$

Notam $x_1 = z \Rightarrow \dot{x}_1 = \dot{z} = x_2$ $\quad \mathcal{L}[x] = X(s)$

$\dot{x}_2 = \ddot{z} = u - 4\dot{z} - 3z = u - 4 \cdot x_2 - 3x_1$

(1): $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} [u]$

$Y(s) = (s+2)Z(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} y = \dot{z} + 2z = x_2 + 2 \cdot x_1$

(2): $y = \underbrace{[2 \quad 1]}_{C} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{[0]}_{D} [u]$

c) $H(s) = \dfrac{(s+2)}{(s+1)(s+3)} = \dfrac{\overset{s+3}{\overset{\diagup}{a}}}{s+1} + \dfrac{\overset{s+1}{\overset{\diagup}{b}}}{s+3} = \dfrac{as+3a+bs+b}{(s+1)(s+3)}$

poli $\{-3; -1\}$

$= \dfrac{(a+b)s + 3a + b}{(s+1)(s+3)}$

$\begin{cases} 1 = a+b \\ 2 = 3a+b \end{cases}$

$1 = 2a \Rightarrow a = \frac{1}{2}$

$b = 1 - a = \frac{1}{2}$.

$\Rightarrow H(s) = \frac{1}{2} \cdot \frac{1}{s+1} + \frac{1}{2} \cdot \frac{1}{s+3} = \frac{Y(s)}{U(s)}$

$\Rightarrow Y(s) = \frac{1}{2} \underbrace{\frac{U(s)}{s+1}}_{X_1(s)} + \frac{1}{2} \underbrace{\frac{U(s)}{s+3}}_{X_2(s)}$

$\mathcal{L}^{-1}$ $\quad \mathcal{L}^{-1}$

$\begin{cases} \dot{x} = Ax + Bu \quad (1) \\ y = Cx + Du \quad (2) \end{cases}$

Notam: $X_1(s) = \dfrac{U(s)}{s+1} \Rightarrow (s+1)X_1(s) = U(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} \dot{x}_1 + x_1 = u$

$\Rightarrow \boxed{\dot{x}_1 = -x_1 + u}$

$X_2(s) = \dfrac{U(s)}{s+3} \Rightarrow (s+3)X_2(s) = U(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} \dot{x}_2 + 3 \cdot x_2 = u$

$\Rightarrow \boxed{\dot{x}_2 = -3x_2 + u}$

(1): $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix}}_{A} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{B} [u]$

$Y(s) = \frac{1}{2} X_1(s) + \frac{1}{2} X_2(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} y = \frac{1}{2} x_1 + \frac{1}{2} x_2$

(2): $y = \underbrace{\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}}_{C} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{[0]}_{D} [u]$

Pr3 Se dă fct de transfer.

$H(s) = \frac{1}{s^2(s+1)}$ Se cere o repr. în SS care să pună în evidență polii sistemului

Structura sistemului:

* Ordin sistem = 3 (grad polinom numitor) $\Rightarrow$ 3 var de stare (x_1, x_2, x_3)

$\Rightarrow$ poli $\begin{matrix} -1 \\ 0 \\ 0 \end{matrix}$ $\Rightarrow$ matrice $A \rightarrow (3 \times 3)$

1 intrare $\Rightarrow$ B $\rightarrow$ 1 coloană

1 ieșire $\Rightarrow$ C $\rightarrow$ 1 linie

$H(s) = \frac{a}{s} + \frac{b}{s^2} + \frac{c}{s+1} = -\frac{1}{s} + \frac{1}{s^2} + \frac{1}{s+1} = \frac{Y(s)}{U(s)}$

$\Rightarrow Y(s) = -\underbrace{\frac{U(s)}{s}}_{X_1(s)} + \underbrace{\frac{U(s)}{s^2}}_{X_2(s)} + \underbrace{\frac{U(s)}{s+1}}_{X_3(s)}$

Notăm $X_1(s) = \frac{U(s)}{s} \Rightarrow s \cdot X_1(s) = U(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} \boxed{\dot{x}_1 = u}$

$X_2(s) = \frac{U(s)}{s^2} \Rightarrow \left(s^2 X_2(s) = U(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} \ddot{x}_2 = u \overset{Nu}{=} \right)$

$$X_2(s) = \frac{\frac{U(s)}{s}}{s} \cdot X_1(s) = \frac{X_1(s)}{s} \Rightarrow s \cdot X_2(s) = X_1(s)$$

$$\xrightarrow{\mathcal{L}^{-1}} \boxed{\dot{x}_2 = x_1}$$

$$X_3(s) = \frac{U(s)}{s+1} \Rightarrow (s+1) X_3(s) = U(s) \xrightarrow{\mathcal{L}^{-1}} \dot{x}_3 + x_3 = u$$

$$\Rightarrow \boxed{\dot{x}_3 = -x_3 + u}$$

$$(1): \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}_{B} [u]$$

$$Y(s) = -X_1(s) + X_2(s) + X_3(s) \xrightarrow{\mathcal{L}^{-1}} y = -x_1 + x_2 + x_3$$

$$(2): y = \underbrace{[-1 \quad 1 \quad 1]}_{C} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{[0]}_{D} [u]$$

Matrice controlabilitate:

$$R_c = [B \quad AB \quad A^2B \cdots A^{n-1}B]$$

$$n = 3 \Rightarrow R_c = [B \quad AB \quad A^2B] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

$B \quad AB \quad A^2B$

$A^2B = A \cdot (AB)$

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{vmatrix} = 1 \neq 0 \Rightarrow \operatorname{rang} R_c = 3.$$

$\Rightarrow$ sistem controlabil

Matricea de observabilitate:

$$R_o = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \overset{n=3}{=} \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \} C \\ \} CA \\ \} CA^2 = (CA) \cdot A \end{matrix}$$

$$\begin{vmatrix} -1 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{vmatrix} = -1 \neq 0 \Rightarrow \text{Rang } O = 3 \Rightarrow \text{sistem observabil}$$

Stabilitate externă

Poli $\in \{-1; 0; 0\} \not\subset \mathbb{C}^- \Rightarrow$ sistem NU este stabil extern

$$[sI - A] = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ -1 & s & 0 \\ 0 & 0 & s+1 \end{bmatrix}$$

$|sI - A| = s \cdot s \cdot (s+1) = 0 \Rightarrow \nabla_A \in \{-1; 0; 0\}$

$\Rightarrow$ sistem la limita de stabilitate d. pd. v intern

pt că $-1 \in \mathbb{C}^-$

$\{0; 0\}$ sunt pe frontiera domeniului de stabilitate

